

Forecasting iBCI decoder failure — clocks, a linear Bayes panel, and a map G

Motor-first project sketch. Start with a physical taxonomy of drift; put three clocks (seconds / minutes / day) on a typed sensor panel; fuse with a hierarchical Bayesian GLM so coefficients are interpretable; optionally learn a class-conditional transport G. A reading list sits at the end of this note (MINDFUL, DREDge, regularized horseshoe first).

0. Start with a physical / literature taxonomy of drift

Before an ensemble, write down what kinds of drift exist in extracellular electrophysiology, on what timescales, and what they do to a frozen decoder. Chronic iBCI papers treat nonstationarity as a fact; they rarely type it. The taxonomy is the first deliverable.

Class	Physical story	Signature	Horizon	Right move
Motion / registration	Probe creeps in tissue (DREDge). Units seem to die and reappear down the shank.	Coherent waveform shift along depth; motion trace.	Seconds–minutes (onset); slow creep over a session	Remap sites / unit IDs. Do not fine-tune a dead slot.
Channel death / isolation	Gliosis, encapsulation, broken trace, bad threshold, unit left the site.	Rate to 0, SNR collapse, impedance open/short, waveform gone.	Hours–weeks (scar); sudden if mechanical	Mask the channel. Recalibrating it invents neighbors.
Reference / artifact	Cable, pump, common-mode, grounding, stimulation.	Many channels jump together; monopolar common-mode.	Seconds	Discard the segment. Never train on it.
State (arousal / fatigue)	Neuromodulatory and behavioral state; geometry unchanged.	LFP low-freq / beta; population rate scale.	Minutes	Gate or add a state feature. Do not rebuild the head on tired.
Code warp	Same units; map to kinematics/phones rotated or gained.	Factor-angle vs rest; preferred-direction walk; soft R-squared fade.	Minutes–days	Small adapter / stitch / the map G below.
Decoder / task confound	Inner speech, rest vs attempt, endpointing, LM lock-in.	Features look fine; posteriors collapse or LM surprisal explodes.	Utterance	Do not touch the array model. Fix the gate.

Motor reaching understates harm because the user closes the loop on a low-D cursor. Speech cannot lean. Use multi-day motor data as the typed testbed; keep speech as the motivation where a miss is a lie rather than a sluggish mouse.

1. Three clocks — levels and derivatives

Predictability is not binary. Onset and ramping faults are differentiable. Step faults (cable yank, unit death at 14:00) are not. Put the same sensors on three clocks. Derivative means a slow difference, not raw d/dt of one LFP sample.

Clock	Feature	Use
vs yesterday / last good day	Level: factor angle, SNR, rate, LFP state vs a reference epoch.	Hours–days. Warp and scar. This session is a different brain.
vs 5–20 min ago	First difference of LFP band power, spike-band gain, isolation, angle.	Fatigue, gain walk, isolation getting messy. Started 5 minutes ago and still going.
vs 10–30 s ago	First difference of DREDge position, artifact RMS, LFP state.	Onset of motion / artifact / sleep-spin in this block.

A second difference is legal only after heavy smoothing: is the fade accelerating? Single-channel second derivatives will fit line noise. Use population summaries.

LFP falling fast is a state/effort alarm, not array death. Compare task-aligned LFP to rest LFP. If only task LFP dies, that is performance. If rest LFP and SNR die together, that is the implant. Combine clocks: LFP slope falling and SNR flat means gate, do not stitch. LFP flat and SNR falling means mask or G.

Only some failures have a forecast horizon. An ensemble that predicts problems is easy to fake with leakage. Class + severity + lead time that physics allows is the paper. If the headline is collapse 6 hours early, reviewers will ask whether you fit time of

day. Split by session and by animal.

Horizon	Plausible	Not plausible
1–30 s	Motion onset (DREDge velocity), artifact, sleep-spin LFP, rate transient	Gliositis; unit will die at 14:00
1–20 min	Fatigue / arousal, spike-band gain walk, isolation getting messy, strategy change	Sudden pull-out
Hours–days	Slow SNR fade, accumulating dropouts, impedance creep, manifold angle wandering	A cable yank

Labels

For every window t , define future decode health on h in {10 s, 2 min, 30 min, rest of day}: delta R-squared (or bits/s, or PER); class (motion / dropout / state / warp / artifact / none); severity bins (none / mild / crash).

Class comes from sensors + injected lesions + lab notes. Severity comes from a frozen reference decoder (day 1, or last-good adapter). Do not use a decoder already updated on the future. That is the leak.

Panel

Features at t , past only: channel SNR / rate / impedance, fraction dead, DREDge position and velocity, unit-match survival, factor-angle vs baseline rest, rest-only noise correlations, LFP state — each as level and short difference. Plus recent decode entropy (past only).

2. Fuse with a linear / GLM Bayes model, not a net

Keep the fusion linear (or a GLM) for as long as it holds. A GBM that works will not tell you whether LFP slope and SNR slope are two names for one thing. The interesting observation is which clock carries lead time, with uncertainty.

Skeleton: a link function of expected y at time t and horizon h equals an animal-and-day intercept plus levels times beta plus short derivatives times gamma plus a few predeclared interactions. y is next-horizon delta R-squared, a crash indicator, or ordinal severity. Predeclare interactions such as LFP-slope times SNR level, motion velocity times unit-survival, angle times rest-only correlation.

Gaussian for delta R-squared; logistic or ordered logit for crash / severity. Regularized horseshoe on the coefficients so the panel can be wide and most weights shrink.

What you can then say

- Posterior that 2-min LFP slope predicts mild loss even when SNR is flat (state, not death).
- Posterior that SNR level eats the long horizon and derivatives add nothing there.
- Whether LFP-slope times SNR is a real interaction or two collinear ramps.

Bayes also gives credible intervals on each clock's lead-time value; partial pooling across animals; proper scoring versus always-recal; a decision threshold on $P(\text{crash given } h)$ from an explicit loss (false alarm vs missed crash).

Cheap expert rules still belong as priors or competing models (DREDge velocity above a threshold means motion). If residuals after the GLM still hide rare conjunctions, put a tree on the residuals or a second-stage class head. Keep the severity head linear. Do not start with a net if the first figure you want is a coefficient table.

Evaluation that is not marketing

- Session-out and animal-out.
- Lead-time distribution given that a crash happened.
- False alarms per hour. 90% recall that beeps every three minutes is useless.
- Policy value: mask / stitch / adapter / pause vs always-recal vs never-recal on next-block R-squared.
- MINDFUL scalar as a nested model. LOO / likelihood ratio against the full panel is figure 1.

Data recipe (EU + public)

1. Multi-day monkey reaching (brainsets / DANDI) — natural drift.
2. Neuropixels + DREDge — motion class with ground truth.

3. Replay + scheduled lesions at known times (stopwatch for lead time).
4. Second animal or human grid = generalization, not the training set.

Infinite compute is for lesion schedules, frozen-decoder seeds, and horizons — not a foundation model on the panel.

Pitch. Levels and short differences of a physically typed panel, in a hierarchical Bayesian GLM with a few interactions, forecast motor-decoder delta R-squared and fault class — and measure which clock actually carries lead time. A class-conditional G is optional and must not be a second foundation model.

3. Optional third arm: a transport G so B looks like A

If at time A a frozen decoder f works and at B it does not (or the monitor predicts it will not), ask whether there is G on neural inputs with $G(x_B)$ approximately x_A in the sense that $f(G(x_B))$ matches the decoder you would have with labels at B. Make today's features look like the day the decoder was honest, without pretending a dead electrode is alive.

The slogan is old (Procrustes, CCA, Cycle-GAN, day-specific layers, POYO session tokens). What is thin is to condition G on the typed alarm rather than always learning a global warp.

Alarm class	Form of G	What not to do
Motion	Permutation / interpolation along the shank (register to A's sites).	A dense MLP that fills in moved units from nowhere.
Dropout / death	Mask those coordinates; G is a projection. Maybe impute from surviving factors.	Force $G(x_B)=x_A$ on a flatline channel.
State (fatigue)	Gain / style on rates or LFP; or concatenate a state bit and leave f alone.	Learn a phone-space warp from tired reaching.
Code warp	Linear G on factors (orthogonal Procrustes, CCA) or a tiny adapter.	A huge unpaired GAN if a 20-D rotation already explains it.
Artifact	G is drop the window. No map.	Any learned G fit on pump noise.

G is underdetermined from unpaired x_A , x_B . Use, in order: (i) known lesion / DREDge as supervision for motion-G; (ii) a few labeled anchor trials at B to pin a linear adapter; (iii) unpaired matching on rest factors only, so the task itself is not the shift you erase. Evaluate f composed with G on next-block R-squared versus retraining f and versus doing nothing.

4. Prior work: not a blank continent

The exact object (forecast + class + policy + typed G) is still thin. Pieces exist. Do not claim nobody thought about drift.

Typical papers either adapt the decoder (manifold alignment, Cycle-GAN, NoMAD, day-specific layers) or measure instability now and correlate it with error. Closest hit: MINDFUL (Pun et al., Commun. Biol. 2024) — label-free score on BrainGate cursor data, r about 0.7–0.9 with closed-loop error over weeks. Awareness, contemporaneous. Not class, not horizon heads, not which surgery, not DREDge + LFP slope + unit-match as a panel.

Say: instability metrics exist; they are contemporaneous and untyped. Ask whether a typed panel can forecast decode failure at physically allowed horizons and select a minimal update. They mostly tried to eat drift with a fancier decoder instead of see it coming.

5. Reading list

If you only read three things: MINDFUL, DREDge, regularized horseshoe + hierarchical intercepts.

The iBCI drift problem

Pun et al., MINDFUL, Communications Biology 2024. Code: github.com/ewinapun/MINDFUL. Beat it on specificity, not on noticing drift.

NoMAD, Nature Communications 2025 — unsupervised stitch so a frozen decoder keeps working. That is G without a forecast.

Farshchian et al., adversarial alignment over days (eLife). Degenhart / Sadtler manifold alignment line.

Front. Comput. Neurosci. 2023 simulation: mean rate / unit count / preferred-direction walk vs OLE-KF-RNN. Use as a lesion catalog.

Willett speech and BrainGate cursor papers: how they actually recalibrate (rolling z-score, day-specific layer).

Physical sensors for the panel

Windolf et al., DREDge, Nature Methods 2025 — motion from AP and LFP. SpikeInterface LFP-drift tutorial. This is x-prime.
 Kilosort / UnitMatch / Bombcell quality metrics — death vs isolation.
 Gallego, Perich, Miller — manifolds; what a factor angle vs yesterday means.
 IBL / Steinmetz Neuropixels; DANDI; brainsets (Perich–Miller) — multi-day files.

Bayesian linear toolkit

Gelman et al., Bayesian Data Analysis, or Regression and Other Stories — hierarchical intercepts by animal/day.
 Carvalho, Polson, Scott — horseshoe. Piironen and Vehtari 2017 — regularized horseshoe. Fit in brms / Stan.
 Vehtari, Gelman, Gabry — PSIS-LOO and calibration. McElreath, Statistical Rethinking, for intuition.
 Short Bayesian decision-theory note: threshold on $P(\text{crash given } h)$ from false-alarm vs miss loss.

Forecast evaluation

Gneiting and Katzfuss — probabilistic forecast evaluation. Reliability diagrams (meteorology papers are clearer than ML blogs).
 Calibration of Bayesian neural decoders (Wei / Pillow-adjacent eNeuro 2024): posteriors lie; $P(\text{crash})$ will too.

Practice

brms + regularized horseshoe on a toy: $y = \text{delta } R\text{-squared}$; predictors = level + 2-min slope of SNR and LFP.
 Replay one Perich animal, inject a dropout at a known time, watch whether the slope coefficient posterior moves before $R\text{-squared}$ dies.
 MINDFUL scalar as a nested model (one feature). That comparison is the first figure.

Framing: instability metrics exist and are untyped. A physically typed panel on three clocks, fused by a hierarchical Bayesian GLM, plus an optional class-conditional G , evaluated as a policy against daily full recalibration on public multi-day motor electrophysiology, is the increment.